Module 12
Business Finance

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION A – CASE QUESTIONS (Total: 50 marks)

Answer 1

Memo

To: Board of Directors of Protex Group Limited
From: Financial Controller
Date: xx/xx/xxxx
Subject: Application of value chain analysis to review and evaluate (i) the value adding activities in mask production and (ii) the potential contribution to the manufacture of new medical production suits

The value chain analysis should be used to review and evaluate activities on whether they add value in the production of medical masks in the Company. The value chain analysis is divided into: (i) primary activities that are directly related to production, sales, marketing, delivery and service – namely inbound logistics, operations, outbound logistics, marketing and sales and after sales services; and (ii) support activities, including procurement, technology development, human resource management and firm infrastructure, which provide supporting functions to make sure primary activities are performed effectively and efficiently and add value to the manufacturing and selling of medical masks.

Inbound logistics activities include receiving, handling and storing raw materials for the manufacturing system. The long-term contracts with suppliers and well-controlled warehouses can ensure that there are sufficient raw materials stored for production. This can help the Company to store and secure raw materials to develop new medical protection suits.

Operations activities relate to the use of raw materials and direct labour to produce medical masks. Since the manufacturing processes of making medical masks are fully automated with limited manual operations, this can avoid manual mistakes in the production processes. The Company has successfully produced high quality medical masks for years, and this experience could be modified to produce new medical protection suits. Nevertheless, the Company may need to acquire new equipment to produce new medical protection suits.

Outbound logistics activities include storage of medical masks in the warehouse and the distribution of medical masks to hospitals in Hong Kong and other Asian markets. The Company has sterilised, secured and well-controlled warehouses to keep all medical masks before selling to customers. The Company also has an experienced sales administration team to arrange shipment and delivery to hospitals in Hong Kong and other Asian markets. These will all create value to the Company to store and distribute new medical protection suits.

Marketing and sales activities are related to advertising, promoting and selling medical masks to hospitals. The Company has an experienced sales team led by a regional sales director to take care of selling activities to hospitals. The success of the Company has proved that the sales activities are performed satisfactorily. These activities can also contribute to the marketing and selling of new medical protection suits to customers. However, a new sales team may need to be established to sell new medical protection suits.

After sales services activities include customer services for handling hospital complaints and for providing advice and guidance to use different medical masks in hospitals. The Company has established an experienced customer services team led by a customer service director who has been providing continuous support to hospitals. This function can also add value in providing customer services to new medical protection suits. A new customer services team may be needed to form and provide customer service to new medical protection suits.

Procurement activities relate to the purchase of raw materials and production equipment to make medical masks. The Company has an established long-term relationship with reputable and high quality raw materials suppliers that can add value to ensure continuous supply of raw materials to develop new medical protection suits. The Company may also need to find new raw materials suppliers to provide raw materials to make medical protection suits.

Technology development activities include product design and improvement in manufacturing processes. The Company has a research and development team to do research and provide new technology to make medical masks. The Company has certain patents to produce medical masks. Nevertheless, it is not sure whether the research and technology used to make medical masks could be applied to make new medical protection suits.

Human resource management activities include recruiting, training, developing and rewarding people in production, sales and customer service in the medical masks business. The Company has different experts and experienced personnel to produce and sell medical masks and to serve customers satisfactorily. The development of new medical protection suits can leverage these areas of expertise and set up new production, sales and customer services teams to develop, manufacture and sell new medical protection suits.

Firm infrastructure activities relate to planning, finance, quality control, internal control structure and processes to make medical masks. The Company has set up an automatic system to manufacture and sell medical masks to hospitals. The existing system can also provide support to the planning and control in the manufacturing of new medical protection suits.

Upon the implementation of above recommendations, the value chain activities can add value to the development, manufacturing and selling of new medical protection suits.

Answer 2

Mr Song has to determine the facts and related ethical issues. The facts are that there is a tremendous demand for medical masks due to the health hazard in the community, and consumers have to buy medical masks to protect their health. The ethical issue is whether the Company should increase the price to sell medical masks to consumers to earn higher profit and sacrifice corporate social responsibility to the community during the pandemic.

Mr Song has to identify the major principles, rules and values of the decision. He is struggling with integrity, which implies fair dealing and truthfulness and acting responsibly and honestly with consumers, and corporate social responsibility, which implies the Company should not take advantage of the health hazard situation by increasing the prices of medical masks to earn tremendous profit. Based on stakeholder's theory, the Company has to consider the various stakeholders' reactions, such as the government, consumers, suppliers and shareholders, when determining whether the Company has acted irresponsibly. The Company and Mr Song may also lose reputation and respect in the community.

Mr Song has to specify the alternative courses of action. The first option is to use cost plus the original profit margin method to set the selling price by taking into consideration the increase of raw material costs due to escalated demand in the market and benchmarking their new selling price to those of similar products in the market. The second option is to sell the medical masks at ten times the existing price to consumers and earn more profit to benefit the shareholders of the Company.

Mr Song needs to assess the positive and negative consequences for the above alternatives in relation to the short-term and long-term prospects of the Company. If Mr Song decides to use cost plus the original profit margin method to set selling prices to sell their medical masks, they will lose the opportunity to earn higher profit for the shareholders, who may be discontented in short-term, but the Company will earn respect and good reputation from consumers and the community and also can retain loyal consumers and shareholders to the Company in the long run. If Mr Song decides to increase the prices of medical masks by ten times the existing price, the Company can earn higher profit for the shareholders and the share price may increase in the short term. Nevertheless, the Company's share prices and prospects may be adversely affected for acting selfishly without considering social responsibilities to the community and consumers.

The final step is to balance the consequences against primary principles or values to select the alternative that best fits the long-term value creation of the Company. Based on the above analysis, it is suggested that Mr Song should not earn super profit but should adjust the selling price after hedging the possible raw materials and logistic price risks for selling the medical masks to consumers for the long-term benefit and sustainability of the Company. This can also balance the interests of the shareholders, and the increased price is still affordable and reasonable to the community. The Company can also attract socially responsible shareholders to invest in the long-term. Moreover, stakeholders may also respect the Company's decision, which can strengthen their relationships with the Company in the long run.

Answer 3(a)

Since this is the first time the Company has made an investment to acquire a medical protection suits manufacturing factory in the Netherlands, the Company should conduct an overseas due diligence before the acquisition, which should include aspects in relation to commercial, financial, legal and intellectual property matters.

Strategic due diligence should be conducted to include strategic analysis that is used to explore the opportunities and threats that the manufacturing factory is facing in manufacturing and selling medical protection suits in the Netherlands and their existing markets by using PEST analysis and Porter's five forces model. Moreover, strategic analysis should also assess the strengths and weaknesses of the manufacturing plant by using resource analysis and value chain analysis.

Operational due diligence should also be conducted to review the existing operations, such as operational facilities, manufacturing processes and operational risks, of the medical protection suits manufacturing factory in the Netherlands.

Financial due diligence should be performed to assess the financial position, including profitability, liquidity, gearing, taxation, contingencies and capital commitments of the manufacturing factory. Particularly, the Netherlands may use different accounting rules to value and measure intangible assets such as patents. These should be reviewed and evaluated because they may significantly affect the acquisition price of the manufacturing factory. The Company should also assess other financial risks, such as credit risk, in the Netherlands that may affect the financial performance of the manufacturing factory and thus should be taken into consideration when setting the acquisition price.

Legal due diligence should be conducted to understand the corporate legal structure and rights of the manufacturing factory in the Netherlands that are new to the Company. The Company should also look at the acquisition laws related to overseas acquisition of a manufacturing factory in the Netherlands to make sure the Company is complying with them during the acquisition.

Intellectual property due diligence should be conducted to investigate the nature and details of the patents and licence to manufacture medical protection suits that the manufacturing factory possesses. These patents and licence are valuable intangible assets in the acquisition, so the Company has to review and look at all patent documents and registration details diligently.

Taxation due diligence should be conducted to evaluate tax issues, such as withholding tax and transfer pricing, and the tax consequence of the acquisition of the medical protection suits manufacturing factory in the Netherlands.

Answer 3(b)

A Netherlands business strategist or consultant who is familiar with the medical protection suits manufacturing industry should be employed to conduct commercial due diligence with the Company's management team.

A Netherlands certified public accountant who is familiar with the financial reporting and risks of the Netherlands should be appointed to work with the finance team of the Company to conduct financial due diligence of the manufacturing factory.

A Netherlands tax specialist should be employed to advise on the most appropriate tax structure for the acquisition with regard to the existing group structure of Protex Group Limited.

A Netherlands lawyer who is familiar with local acquisition laws should be employed to work with a Hong Kong lawyer appointed by the Company to conduct legal due diligence.

A Netherlands intellectual property lawyer who is familiar with intellectual property laws should work with a Hong Kong intellectual property lawyer appointed by the Company to conduct intellectual property due diligence.

A Netherlands intellectual property valuation expert should also be appointed to assess the market value of intellectual property possessed by the manufacturing factory based on specific local business and industry information.

Answer 4

The most appropriate method to assess the financial feasibility of Project Adventure is the net present value method using nominal values based on the preference of the Company.

Year	0	1	2	3	4
	US$	US$	US$	US$	US$
Investment	(1,000,000)				
Sales (Note 1)		1,020,000	1,404,000	1,590,000	1,620,000
Costs of sales (30%)		(306,000)	(421,200)	(477,000)	(486,000)
Gross profit		714,000	982,800	1,113,000	1,134,000
Operating expenses (Note 2)		(550,800)	(624,000)	(699,600)	(712,800)
Depreciation		(125,000)	(125,000)	(125,000)	(125,000)
Profit before tax		38,200	233,800	288,400	296,200
Tax (30%)		(11,460)	(70,140)	(86,520)	(88,860)
Profit after tax		26,740	163,660	201,880	207,340
Add: Depreciation		125,000	125,000	125,000	125,000
Net cash flow	(1,000,000)	151,740	288,660	326,880	332,340
Cash flow from disposal (Note 3)					604,621
	(1,000,000)	151,740	288,660	326,880	936,961
Discount factor (Note 4)	1.00	0.85	0.72	0.61	0.52
Net Present value	(1,000,000)	128,979	207,835	199,397	487,220

(Note 1)

	1	2	3	4
Sales (real)	1,000,000	1,350,000	1,500,000	1,500,000
Inflation (2%)	1.02	1.04	1.06	1.08
	1,020,000	1,404,000	1,590,000	1,620,000

(Note 2)

	1	2	3	4
Operating expenses (real)	540,000	600,000	660,000	660,000
Inflation (2%)	1.02	1.04	1.06	1.08
	550,800	624,000	699,600	712,800

(Note 3)

Proceeds in year 4 (nominal)	649,459	649,459
Less: Net book value of the assets	(500,000)	
Gain on disposal before tax	149,459	
Less: Tax on the disposal gain (30%)		(44,838)
Cash flow from disposal		604,621

(Note 4)

Risk premium	Probability	
Prosperous economy (6%)	50%	3%
Poor economy (10%)	50%	5%
		8%

Fisher equation:

Nominal Interest rate = [(1 + 0.08)(1 + 0.02) - 1] x 100%
= 10.16% ≈ 10%

Discount factor 18%

Since the net present value is US$23,431 and positive, the Company should invest in Project Adventure.

* * * END OF SECTION A * * *

SECTION B – ESSAY / SHORT QUESTIONS (Total: 50 marks)

Answer 5(a)(i)

	Forecast July HK$'000	Forecast August HK$'000
Service fee receipts	650	3,150
Office rent	180	180
Salaries and MPF to full-time staff	240	270
Salaries to temporary workers	65	110
Performance fees of pop singers	600	430
Decoration materials costs	567	1,625
Delivery and installation costs	158	325
Rental of equipments	35	221
Total disbursement paid	1,845	3,161
Net cash outflow	(1,195)	(11)
Cash balances brought forward	2,000	805
Forecasted cash balances at the end of the month	805	794

Answer 5(a)(ii)

Working capital needed for August operation
HK$3,161,000

Extra funding required to sustain August operation
HK$3,161,000 – HK$805,000
= HK$2,356,000

Working capital required to sustain August operation if the Company postpones the projects in August (HK$'000)
180 + 270 + (110 x 50%) +221
= 726

HK$805,000 is able to cover HK$726,000, so no further funding is required.

Answer 5(b)

The Company can consider centralising the procurement of decoration materials, delivery and installation by consolidating their procurement into a few authorised suppliers and negotiating a credit term for their purchases. Besides better pricing, consolidating their procurement allows the Company to have a steadier and larger scale of purchase to negotiate for better terms with their suppliers. The Company can negotiate with its suppliers for an open account which issues a monthly statement instead of paying its suppliers immediately by cash.

As the customers seem to settle the invoices immediately without asking for longer credit terms, the Company shall issue the invoices to their customers immediately after the events instead of waiting till the end of the month. The current practice will impact the working capital of the Company, especially for those events that are completed at the beginning of the month, as the Company has to wait for almost one month before receiving the cash.

The Company can also negotiate with its customers to place a portion of the fees as deposits in advance before the event. Placing deposits is a very common practice which can enhance the liquidity of the Company.

Answer 5(c)

The dividend policy is a key part of the long-term financial strategy pursued by the management and is a major part of the Company's relationships with its shareholders. Retained earnings is an internal source of cheap funds for financing the growth of the Company as well as funding the working capital of the Company, whilst dividends, from the shareholders' point of view, are a source of income or return on investment.

The payment of dividends is a cash outflow from the Company. Once the dividend is paid, it cannot be recalled by the Company if it subsequently finds itself in an illiquid position. The Company therefore needs to ensure it has sufficient liquidity to pay declared dividends and also to meet its liquidity obligations required for continuing funding and investment decisions. Using external debts to finance working capital involves interests, which affects the profitability of the Company at the same time.

For a private company, the shareholders will compare the interest rate on loans borrowed by the Company with the opportunity cost of the fund, i.e. the investment returns of the fund to be distributed to them as dividends. However, if the Company consists of more than one shareholder, the opportunity cost of the fund could be different, so the preference would vary.

The Company is recommended to develop a long-term and consistent dividend policy which can balance the expectation of shareholders and the needs of the Company.

As the forecasted cash position is below HK$1 million at the end of August, it is not recommended to borrow to fund the dividend payment. However, developing a banking relationship by having a bank loan facility may help the Company seek financing when management has an opportunity to grow the business.

Answer 6(a)

Current ratio

2021:	11,406 / 8,422	= 1.35
2020:	8,804 / 6,603	= 1.33

Yes, the current ratio remained steady and even slightly increased from 1.33 to 1.35.

<u>Liquidity</u>

Quick ratio

2021:	(4,598 + 3,663) / 8,422	= 0.98
2020:	(4,199 + 2,850) / 6,603	= 1.07

or

Current assets to total liabilities

2021:	11,406 / 17,776	= 0.64
2020:	8,804 / 12,728	= 0.69

or

Working capital to total assets

2021:	(11,406 – 8,422) / (11,406 + 29,751)	= 0.07
2020:	(8,804 – 6,603) / (8,804 + 18,976)	= 0.08

<u>Profitability</u>

Gross profit margin

2021:	5,383 / 17,258	= 31.19%
2020:	5,875 / 16,014	= 36.69%

Although the current ratio is improved slightly, the quick ratio/ current assets to total liabilities/ working capital to total assets is getting worse. This indicates the Company's short-term liquidity position is almost unable to meet its obligations.

The gross profit margin decreased by 5.5%, indicating the Company was under price pressure and might have reduced their selling price to sustain their sales growth. The cost of goods sold could also have increased, giving cause to the lower gross profit margin.

Inventories turnover days

2021:	2,645 / 11,875 x 365	= 81.30
2020:	1,355 / 10,139 x 365	= 48.78

Account receivables turnover days

2021:	3,663 / 17,258 x 365	= 77.47
2020:	2,850 / 16,014 x 365	= 64.96

The increase in inventories turnover days from 48.78 to 81.30 might indicate a significant increase in inventory and not enough sales growth year on year, and there is also a risk of declining demand for their products. This might also indicate that the Company was not managing its inventories properly, and there could be a potential increase in obsolete inventories.

Accounts receivables turnover days was also increased, which indicated customers were repaying the Company slower or that there might even be overdues. This indicates that customers' ability to pay has deteriorated and may need to consider the adequacy of the provision for doubtful debt.

Answer 6(b)

In addition to the financial measures adopted by the Company to measure performance, the Company shall also take measures to enforce their strategic control to improve performance. Strategic control helps the Company to revisit the assumptions underlying the current strategy during the implementation stage, continue to monitor and evaluate the strategic milestones, and take any necessary corrective actions. Strategic control represents the processes to monitor and control the execution of a strategic plan for assessing long-term issues.

- Has the strategy been implemented as planned?
  Measure the success of the R&D, new patents and products launched.

- Are the assumptions about major trends and changes correct?
  Compare the products and equipments with competitors in the markets.
  Benchmark against the competitors.

- Are objectives and schedules being met?
  Measure the actual timeline for new product launches and compare with their planned timeline.

- Are revenues, costs, and cash flows supported by the assumptions and matched with the projections?
  Explain reasons and root cause for major deviations.

- Are any operational changes necessary? When managing inventories, will the just in time or bill to order model be used?

Other valid point

Answer 7(a)

Managing an organisation's banking relationships constitutes one of the key parts of the treasury department's responsibilities. Maintaining a mutually beneficial and transparent relationship between the organisation and the bank is a critical factor towards achieving access to various treasury functions such as providing funding, processing payments, arranging hedging, and liquidity management. Banking relationships are normally managed by the relationship manager of the bank who is assigned to the company, and they normally provide these varied services and advice through their in-house product specialist. Banks often provide informal advice to the treasury department.

In this case, since the Company is exposed to various currencies such as HKD, RMB, USD and EUR, the Company could request the bank to provide hedging advisory services with foreign exchange products such as forwards and options. As the Company is selling to their distributors with credit, one of the solutions to working capital management could be factoring (or forfaiting) which banks are active participants. Factoring (or forfaiting) is the discounted sale of a receivable. Banks provide important funding as well as reduce credit risk for the customer.

Since the Company does not have any debt financing, the Company can consider inviting the bank to provide debt issuance and related advisory services. The bank can either assist the Company in issuing commercial bonds or provide a bank loan to the Company to finance the project or the growth of the Company.

Answer 7(b)

A company that issues debt can elect to have its debt rated by a credit rating provider. Since Nicole believes the Company's financial results are positive, strong cooperation between the Company and the credit rating agency can provide a positive message to the Company's bond investors.

The treasurer as well as the treasury team will be the key contacts of the credit rating analysts. They need to respond to the information requests from the credit rating provider's review team. The credit rating would be assigned to a company based on the information provided.

Generally, the analysts will ascribe about 50% of the rating to the Company's business profile and future prospects and 50% to its current financial profile. The analyst team uses financial statements publicly available. The analyst team will also assess forecasts and budgets, operating performance, risks and risk management approaches and the competitive landscape in the market. The analysis sometimes will start from a questionnaire, several years of historical financial data, forecasts, a business overview, a background of the senior executives, and a snapshot of the competitive landscape. The ratings focus is on forecasting the Company's performance into the future, rather than considering past performance.

The credit rating agency sometimes may request the treasurer to conduct a presentation, or arrange an interview with the Company's senior management, to let the credit rating team to question specific areas or items of concerns to assess the Company's financial stability and the likelihood of achieving its strategic objectives.

Answer 7(c)

The Company can complain to the bank for misrepresenting that the structure products in Australian dollars is good for hedging the foreign currency exposure of the Company as the Company did not have any currency exposure in Australian dollars, so there is no necessity for the Company to hedge the Australian dollar.
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The Company can also complain to the Hong Kong Monetary Authority ("HKMA"), as HKMA promotes the stability and integrity of the financial system, including the banking system, and helps to maintain Hong Kong's status as an international financial center. HKMA is also the regulator of the Banking Ordinance. In this case, the bank was making an inducement to buy a structured product by providing a misleading rationale, i.e. to place the structure deposits in Australian dollars for hedging purposes, which potentially violates the guidelines on business practices of authorised institutions published by HKMA. The Hong Kong Monetary Authority is the frontline regulator of these activities.

The Company can also complain to the Securities and Futures Commission ("SFC"). The SFC requires intermediaries who deal and advise in futures contracts to register with the SFC. As the relationship manager is not registered with the SFC, he has violated the Securities and Futures Ordinance in dealing and advising in futures contracts. Also, as the SFC reviews offering documents and features of structured investment products including structured deposits, the Company can complain if the functions of these two products were not properly disclosed when the bank was recommending these to the Company.

Answer 7(d)

The Company is recommended to implement a risk management control system including preventing, detecting, and mitigating potential risks in the following areas.

<u>Governance structure</u>

The Company shall develop a governance structure to provide oversight and accountability for actions. The Board of Directors provides overall oversight and directions for the Company, and the treasury can provide professional advice on financial risks to the CFO and Board of Directors. The treasury department can also assist the Company in setting up a Risk Management Committee and can be one of the members who specialises in financial risks and formulate objectives regarding financial risk management.

The Risk Management Committee can report directly to either the Board or Audit Committee.

Control Environment

A comprehensive control framework, from oversight and supervision to specific process controls, can help the Company to achieve its business objectives and to mitigate and deter unethical conduct.

The treasury department can help to develop standards, policies and procedures, and communicate properly to the relevant staff of the Company. Clear and documented roles and responsibilities throughout the Company are critical to ensuring employees perform their functions in accordance with management expectations. The treasury department can also review business processes, such as management reporting, process documentation, technology and systems controls, or approval requirements, and suggest controls which can prevent or limit the opportunity for fraud and other unethical behavior related to financial risks.

The treasury department can advise on payment approval process and control, the appropriate delegation of authority for payment, the credit terms approval process to customers, and the execution of the credit policy to customers, i.e. dealing with overdue or excess credit limits. The treasury department can also develop investment and hedging guidelines and policies.

* * * END OF EXAMINATION PAPER * * *